Keywords: KV cache compression • mixture-of-experts • Fisher information • efficient LLM inference

MoEZip: Routing-Aware KV Cache Compression for Sparse Mixture-of-Experts LLMs

A Fisher-Information Routing Metric Cuts KV Memory by 10× While Preserving Expert Dispatch—Outperforming the Prior Art by 2.3–4.9×

By Minsang Kim, Seungjun Baek

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Sparse mixture-of-experts (MoE) language models are uniquely vulnerable to KV cache eviction: removing context entries does not merely reduce attention coverage—it changes which experts are selected at each decode step, corrupting generation in ways that attention-only importance scores cannot predict. MoEZip closes this gap with a Fisher-information-based routing sensitivity metric that, combined with instruction-token attention weight, identifies which KV entries can be safely dropped. At a 10% retention ratio, MoEZip surpasses the prior state-of-the-art by +24 and +20 average points on LongBench and RULER—a 2.3× and 4.9× multiplicative improvement—without any model fine-tuning.

AT A GLANCE

Problem: KV eviction in sparse MoE LLMs alters which experts are selected at decode time; attention-only scores are blind to this routing perturbation.

Why it matters: MoE models (Mixtral, DeepSeek-MoE, etc.) are the dominant efficiency paradigm; deploying them at long contexts requires KV compression, yet existing methods degrade generation quality.

Method: Routing sensitivity via per-position Fisher information of the router distribution, combined with instruction-token attention scores, gives a dual-signal importance measure for each KV entry.

Result: At 10% KV retention, MoEZip scores 2.3× higher on LongBench and 4.9× higher on RULER than the prior SoTA (KVZip).

Evidence: Evaluation on LongBench, RULER, and NIAH across both softmax-type and sigmoid-type MoE routing architectures.

The Big Picture

Sparse MoE models reduce per-token floating-point operations by routing each token to a small subset of expert feed-forward networks. The key computational saving is that most parameters remain dormant on any given token. However, memory at inference scales with context length through the KV cache, and reducing it through eviction is essential for long-context deployment.

The standard approach—evict entries with the lowest cumulative attention weight—works well for dense transformers, where every parameter is active for every token. It fails silently in MoE models because the router is a function of the hidden state produced by the attention layer, and the hidden state depends on which KV entries are present. Removing a seemingly low-attention entry can shift the attention distribution enough to change the router's top-k expert selection, propagating an error that compounds across all subsequent layers.

Why Existing Approaches Fall Short

Attention-only scoring (H2O, SnapKV, PyramidKV): These methods score KV entries by the attention weight they accumulate across query tokens. They are designed for dense transformers and have no term that reflects the router's sensitivity to perturbations in the attention output.

Magnitude-based pruning: Low-magnitude entries may reside in positions where the routing gate function is highly sensitive; magnitude and routing sensitivity are not correlated in general.

KVZip: The prior SoTA for MoE KV compression, but its importance score does not account for routing distribution shift caused by eviction, leaving a systematic gap that MoEZip addresses.

Core Mechanism

MoEZip scores each KV entry (i, l) (context position i, layer l) with two signals:

1. **Routing sensitivity $\mathcal{F}_l(i)$:** How much does evicting position i shift the expert selection probabilities at layer l? Derived from the Fisher information of the routing distribution with respect to the attention output at position i.
2. **Instruction attention $\alpha_l(i)$:** The accumulated attention weight that instruction tokens (the task-relevant prefix) assign to position i at layer l.

The final importance score is $\phi_l(i) = \alpha_l(i) + \lambda \cdot \mathcal{F}_l(i)$, with λ balancing the two signals. Entries with the lowest $\phi_l(i)$ are evicted. No gradient computation or additional data is required.

Technical Formulation

Let $p_l(i) = \text{softmax}(W_r h_l(i))$ be the expert routing distribution at layer l, conditioned on the hidden state $h_l(i)$

produced by attending over context position i. The Fisher-information routing sensitivity is:

$$\mathcal{F}_l(i) = \mathbb{E}_{e\sim p_l(i)}\Big[\big(\nabla_{h_l(i)} \log p_l(i)[e]\big)^2\Big]$$

For softmax routing, this simplifies to a closed form involving the variance of the routing logits:

$$\mathcal{F}_l(i) = \mathrm{Var}_{e\sim p_l(i)}\Big[W_r^{(e)} h_l(i)\Big]$$

where $W_r^{(e)}$ is the e-th row of the routing weight matrix. An analogous closed form is derived for sigmoid-type routing. The combined score is:

$$\phi_l(i) = \alpha_l(i) + \lambda \cdot \mathcal{F}_l(i)$$

Entries with the lowest $\phi_l(i)$ values are evicted to meet the retention budget.

Learning or Inference Procedure

MoEZip is entirely training-free. At inference:

1. During the prefill phase, compute $\mathcal{F}_l(i)$ and $\alpha_l(i)$ for each KV position i and layer l.
2. Sort positions by $\phi_l(i)$ in descending order and retain the top-k (where k is the retention budget).
3. Proceed with autoregressive decoding over the compressed KV cache.

No gradient is computed; the Fisher computation reuses quantities already available from the attention forward pass. The method is plug-in compatible with any MoE LLM.

What the Guarantee Says

The Fisher-information derivation provides a local linear approximation to the change in the routing distribution caused by evicting position i. Specifically, the first-order routing perturbation $\Delta p_l(i)$ satisfies:

$$\|\Delta p_l(i)\|_2^2 \approx \mathcal{F}_l(i) \cdot \|\Delta h_l(i)\|_2^2$$

where $\Delta h_l(i)$ is the change in the hidden state caused by removing entry i. Retaining entries in decreasing order of $\mathcal{F}_l(i)$ therefore minimizes the worst-case routing perturbation norm at the given budget.

Experimental Findings

- **LongBench (long-document understanding):** At 10% KV retention, MoEZip scores 2.3× higher than KVZip on the average metric across six long-document subtasks.
- **RULER (structured reasoning):** MoEZip achieves 4.9× higher scores than KVZip at 10% retention; the gains are especially large on multi-hop retrieval subtasks where expert dispatch plays a critical role.
- **NIAH (needle-in-a-haystack):** MoEZip successfully retrieves needles at compression ratios where all attention-only baselines fail.
- **Both routing types:** Results are consistent across softmax-type (Mixtral-style) and sigmoid-type (DeepSeek-style) MoE architectures.
- **High retention convergence:** At $\geq$50% retention, all methods converge toward full-cache performance, confirming the gains are specific to the high-compression regime.

Ablations and Interpretation

Routing-only scoring ($\lambda = 1$, $\alpha = 0$): Underperforms on tasks where instruction-token attention is the dominant signal; confirms both terms are needed.

Attention-only scoring ($\lambda = 0$): Matches existing dense-transformer baselines, confirming that the routing-aware term alone drives the MoEZip gains.

Sensitivity to λ: Performance is robust to $\lambda \in [0.5, 2]$; the default $\lambda = 1$ performs well across all benchmarks.

Computational overhead: The Fisher computation adds $< 5\%$ latency to the prefill phase and is negligible relative to the decode speedup from the compressed KV cache.

Reference

Minsang Kim, Seungjun Baek. **MoEZip: Routing-Aware KV Cache Compression for Sparse Mixture-of-Experts LLMs**. arXiv:2610.30206. NeurIPS 2026. https://arxiv.org/abs/2610.30206

Keywords: text-to-image generation • compositional reasoning • iterative refinement • GRPO

FiRe: Fine-grained Multimodal Reasoning for Enhanced Image Generation

Requirement-Level Check-Refine Cycles and Per-Step RL Rewards Fix the Fix-One-Break-Another Problem in Compositional T2I

By Yongjin Kim, Yoonjin Oh, Yerin Kim, Hyomin Kim, Jeeyoung Yun, Yujung Heo, Minjun Kim, Sungwoong Kim (KU-AGI)

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Text-to-image models fail on compositional prompts because whole-image feedback cannot isolate which individual visual requirement is violated—so fixing one element inadvertently breaks another. FiRe decomposes prompts into atomic visual requirements, checks each one independently using a multimodal LLM, and applies *local* refinements to only the failing regions. FiRe-GRPO, the accompanying reinforcement learning variant, assigns a per-step reward to each check-refine action, enabling the model to learn when and how to refine rather than merely whether the final image is correct. On compositional benchmarks including T2I-CompBench and GenAI-Bench, FiRe achieves consistent improvement over T2I baselines, with the largest gains on prompts specifying three or more simultaneous visual constraints.

AT A GLANCE

Problem: Compositional T2I prompts impose multiple simultaneous visual constraints; whole-image feedback cannot localize failures or guide targeted corrections.

Why it matters: State-of-the-art T2I models still fail on multi-constraint prompts despite improved language understanding; the bottleneck is the feedback signal, not the generator.

Method: Decompose the prompt into visual requirements; check each with an MLLM; refine locally; train with FiRe-GRPO assigning step-level rewards to each check-refine action.

Result: Consistent improvement on compositional benchmarks, especially on prompts with ≥ 3 simultaneous constraints.

Evidence: T2I-CompBench, GenAI-Bench; ablations isolate local vs. global refinement and step-level vs. terminal-only reward.

The Big Picture

Language-guided image generation has improved dramatically, yet compositional prompts—requests that simultaneously specify object counts, attributes, spatial relationships, and interactions—remain a known failure mode. A prompt such as "a red cup to the left of a blue vase on a wooden table" imposes three simultaneous visual constraints; any failure in the generated image requires identifying *which* constraint is violated before attempting a correction.

Existing reasoning-augmented approaches attempt to fix this by elaborating the prompt before generation or by scoring the whole image after generation. Neither approach can localize the failure to a specific requirement or avoid disrupting already-satisfied constraints during refinement.

FiRe reframes image generation as a sequential decision process where each step either checks one requirement or applies one local refinement, and learns via reinforcement learning to execute this process efficiently.

Why Existing Approaches Fall Short

Prompt augmentation (chain-of-thought elaboration): Improves average image quality but generates the same elaborated prompt for every seed; it cannot recover from specific failures in a generated image.

Whole-image feedback (DALL-E Reflect, ImageReward): Produces a scalar or sentence-level critique. This is too coarse to guide targeted corrections and does not indicate which region or property to change.

Terminal-only RL for T2I: Assigning reward only to the final image creates a sparse signal that is difficult to optimize and does not credit or penalize intermediate refinement decisions.

Core Mechanism

Step 1 — Requirement decomposition. An MLLM parses the prompt into a list of atomic visual requirements $\mathcal{R} = \{r_1, \ldots, r_n\}$, where each r_i is a checkable property (e.g., "red cup present", "cup to the left of the vase").

Step 2 — Initial generation. A base T2I model generates I_0 from the prompt.

Step 3 — Fine-grained check. For each unsatisfied requirement r_i, the MLLM answers "Is r_i satisfied in I_{t-1}?" with a binary verdict and a natural-language diagnosis specifying the region and nature of the failure.

Step 4 — Local refinement. The least-satisfied requirement $i^* = \arg\min_i \mathcal{V}_i(I_{t-1})$ triggers an inpainting or latent edit confined to the region described in the diagnosis, yielding I_t. Unaffected regions are left unchanged.

Step 5 — Iteration. Steps 3–4 repeat until all requirements are satisfied or the step budget T is exhausted.

Technical Formulation

Let $\mathcal{V}_i(I) \in [0,1]$ denote the MLLM's estimated probability that requirement r_i is satisfied in image I. The step-level reward at refinement step t (targeting requirement i^*) is:

$$\rho_t = \mathcal{V}_{i^*}(I_t) - \mathcal{V}_{i^*}(I_{t-1})$$

The FiRe-GRPO policy gradient objective is:

$$\mathcal{J}_{\text{GRPO}}(\theta) = \mathbb{E}\left[\sum_{t=1}^{T} \rho_t \cdot \log \pi_\theta(\delta_t \mid I_{t-1}, r_{i^*})\right] - \beta \cdot D_{\text{KL}}(\pi_\theta \| \pi_{\text{ref}})$$

where δ_t is the refinement action (diagnosis + edit), π_θ is the MLLM policy being trained, and π_{ref} is a frozen reference policy. The KL term prevents the policy from diverging far from a well-behaved prior and controls mode collapse.

The combined requirement satisfaction score at the end of generation is:

$$\text{Score}(I_T) = \frac{1}{n} \sum_{i=1}^{n} \mathcal{V}_i(I_T)$$

This is the final metric reported on benchmarks.

Learning or Inference Procedure

Training:

1. Sample a compositional prompt; decompose into requirements.
2. Generate I_0 with the base T2I model.
3. Run the check-refine cycle for T steps; collect $\{\rho_t\}$.
4. Update the MLLM backbone with FiRe-GRPO.
5. The 300-step checkpoint (FiRe-300Step) is used for all evaluations.

Inference:

1. Parse the prompt; generate I_0.
2. Iterate: check all requirements, identify the most violated, apply a local refinement.
3. Return I_T after the budget is exhausted or all requirements are satisfied.

What the Guarantee Says

The step-level reward ρ_t provides a per-requirement improvement signal. Under mild assumptions on the MLLM's calibration (i.e., $\mathcal{V}_i$ monotonically correlates with human satisfaction judgments), the GRPO objective with the KL regularizer converges to a policy that maximizes expected total requirement satisfaction while remaining close to the reference policy. This ensures that FiRe does not sacrifice the reference model's general image quality in pursuit of compositional accuracy.

Experimental Findings

- **T2I-CompBench:** FiRe improves attribute binding and spatial relationship accuracy over all baselines; largest gains on prompts with ≥ 3 simultaneous constraints.
- **GenAI-Bench:** Consistent improvement across compositional categories (color, count, spatial, interaction).
- **Fix-one-break-another:** Global refinement baselines show improvement in the targeted attribute but regression in other attributes; FiRe's local edits preserve already-satisfied requirements.
- **FiRe-GRPO vs. supervised fine-tuning:** Step-level RL converges faster and produces better refinement policies than training with supervised labels for each step.

Ablations and Interpretation

Global vs. local refinement: Applying refinement to the whole image rather than the failing region degrades already-satisfied requirements by an average of 12 percentage points on T2I-CompBench; local refinement avoids this regression entirely.

Step-level vs. terminal-only reward: Terminal-only reward (only $\mathcal{V}_{i^*}(I_T)$ at the final step) trains 40% slower and achieves 8 percentage points lower compositional accuracy; step-level rewards provide a denser credit assignment signal.

Number of refinement steps T: Performance improves with T up to 3–4 steps and plateaus beyond; additional steps add compute without further gains on typical 3-constraint prompts.

Without requirement decomposition: Using a single holistic verdict instead of per-requirement checks degrades performance to below the whole-image feedback baseline, confirming that the decomposition is the key structural contribution.

Reference

Yongjin Kim, Yoonjin Oh, Yerin Kim, Hyomin Kim, Jeeyoung Yun, Yujung Heo, Minjun Kim, Sungwoong Kim. **FiRe: Fine-grained Multimodal Reasoning for Enhanced Image Generation**. arXiv:2604.13491. NeurIPS 2026. https://arxiv.org/abs/2604.13491

Keywords: multimodal LLM evaluation • visual cognition • psychometric benchmark • spatial reasoning

Human Cognitive Benchmarks Reveal Foundational Visual Gaps in MLLMs

VisFactor: 23 Frontier Models Score Below 31% on Psychometric Visual Tests Normed to Human Populations

By Jen-Tse Huang, Dasen Dai, Jen-Yuan Huang, Youliang Yuan, Xiaoyuan Liu, Wenxuan Wang, Wenxiang Jiao, Pinjia He, Zhaopeng Tu, Haodong Duan (CUHK-ARISE)

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Multimodal LLMs score impressively on academic visual benchmarks, yet those benchmarks allow language priors and common-sense shortcuts to compensate for weak raw visual perception. VisFactor digitizes 20 vision-centric subtests from the Factor-Referenced Cognitive Tests (FRCT)—a decades-old psychometric battery designed to isolate specific human visual faculties—and reveals that 23 frontier MLLMs, including GPT, Gemini, and Qwen families, achieve a best score of 30.17 out of 100. The human norm is approximately 80. Neither scale nor chain-of-thought prompting closes the gap, exposing foundational deficits in spatial reasoning and visualization that conventional benchmarks conceal.

AT A GLANCE

Problem: Academic MLLM benchmarks allow language shortcuts and miss the low-level visual cognition that human vision relies on.

Why it matters: Systems deployed in visually demanding roles (medical imaging, robotics, design) require the cognitive visual capabilities that current benchmarks do not probe.

Method: VisFactor digitizes 20 FRCT vision subtests spanning spatial reasoning, perceptual speed, visualization, and visual memory into zero-shot multiple-choice questions.

Result: Best of 23 frontier MLLMs scores 30.17/100; human norm $\approx$ 80; no model reaches the human level on any domain.

Evidence: 23 MLLMs (GPT, Gemini, Claude, LLaMA, Qwen, InternVL); ablations confirm visual engagement, rule out position bias.

The Big Picture

The standard narrative in multimodal AI is one of rapid convergence: each new MLLM generation narrows the gap to human performance on VQA, image captioning, and visual reasoning benchmarks. This narrative is misleading because those benchmarks are designed by ML researchers and systematically admit language-based strategies that bypass genuine visual computation.

Cognitive psychologists have known for decades that human visual cognition is a multi-factor system encompassing spatial visualization, perceptual speed, form completion, and short-term visual memory—each measurable independently with psychometric instruments normed to large human populations. VisFactor applies those instruments to MLLMs and finds a strikingly different picture: all frontier models score far below the human norm on every domain, with the largest gaps in 3D-to-2D spatial transformations and form completion.

Why Existing Benchmarks Fall Short

Language shortcuts: Many VQA questions can be answered partially from the question text alone; models with strong language priors achieve inflated scores without parsing the image carefully.

Coarse granularity: Aggregate accuracy on tasks like MMBench or MMMU merges many heterogeneous subtasks, making it impossible to identify which specific visual process is failing.

Saturation and ceiling effects: Frontier models now saturate several prominent benchmarks, leaving no headroom to distinguish capability levels among top systems.

Lack of psychometric grounding: Existing benchmarks carry no normative calibration against human populations, so "state-of-the-art" accuracy has no reference point against actual human visual cognitive ability.

Core Mechanism: VisFactor

VisFactor digitizes 20 vision-centric subtests from the FRCT battery, grouped into four domains:

Spatial Reasoning: Card Rotations, Paper Folding—tests of mental rotation and spatial visualization in 3D.

Perceptual Speed: Finding A's, Number Comparison—tests of rapid figure-ground segregation and symbol matching.

Visualization: Surface Development, Form Board—tests of 3D-to-2D reasoning and shape completion.

Visual Memory: Shape Memory, Position Memory—tests of short-term retention of visual configurations.

Each item is digitized as a multiple-choice question. The FRCT scoring key provides ground-truth answers. No image captioning or text-only reasoning is possible on any item; answering requires genuine visual computation.

Technical Formulation

Let M be an MLLM, $\mathcal{Q} = \{(q_i, A_i, a_i^*)\}_{i=1}^N$ the question set (image q_i, options A_i, correct answer a_i^*). The VisFactor score is:

$$\text{Score}(M) = \frac{1}{N}\sum_{i=1}^{N} \mathbf{1}[M(q_i, A_i) = a_i^*] \times 100$$

Domain subscores partition the sum over the four cognitive domains. Random baseline performance is 25 (four options). Human FRCT norm is approximately 80. The gap $\Delta(M) = 80 - \text{Score}(M)$ measures how far a model is from average human performance.

Learning or Inference Procedure

VisFactor is a zero-shot evaluation protocol:

1. Present the model with the item image and the four multiple-choice options.
2. Use a fixed instruction: "Which option is correct? Answer with a single letter."
3. Record the model's response; compare to the FRCT scoring key.
4. Compute domain subscores and total score.

A chain-of-thought variant instructs models to explain their reasoning before answering. Both standard and CoT protocols are reported.

What the Guarantee Says

VisFactor's psychometric grounding provides several validity properties:

- **Construct validity:** FRCT subtests were validated against factor-analytic studies of human visual cognition; each subtest measures a theoretically distinct visual factor.
- **Convergent validity:** High within-domain correlations across models confirm that subtests in the same domain measure a common latent visual ability in MLLMs.
- **No language-shortcut confound:** Removing the image reduces all models to chance (25%), confirming that scores reflect visual engagement rather than language priors.

Experimental Findings

- **Best model:** Scores 30.17/100; human norm ≈ 80. All 23 models score between 25 (chance) and 31.
- **Domain asymmetry:** Perceptual speed $\approx 45\%$; spatial reasoning $\approx 28\%$; visualization $\approx 20\%$; visual memory $\approx 22\%$.
- **Scale does not help:** Larger models within each family show no consistent improvement on VisFactor, unlike most academic visual benchmarks.
- **Chain-of-thought:** Marginal improvement on spatial reasoning; slight degradation on perceptual speed (CoT slows the rapid comparison process the subtest is designed to measure).
- **No saturation:** No model approaches the human norm on any domain, leaving substantial room for future progress.

Ablations and Interpretation

Image removal: All models score $\approx 25\%$ when the image is removed and only the question text and options are provided, confirming genuine visual engagement.

Option order shuffling: Results are stable under option permutation (variance < 2 percentage points), ruling out position bias as a confound.

Item difficulty calibration: Items requiring mental rotation of complex 3D figures are the hardest; items requiring simple symbol comparison are easiest—matching the psychometric structure designed for humans. The cross-model difficulty ordering correlates with human item difficulty at $r = 0.73$.

Closed-source vs. open-source models: Closed-source models (GPT, Gemini, Claude) score marginally higher than open-source models of comparable size, but the gap is small relative to the distance from the human norm.

Reference

Jen-Tse Huang, Dasen Dai, Jen-Yuan Huang, Youliang Yuan, Xiaoyuan Liu, Wenxuan Wang, Wenxiang Jiao, Pinjia He, Zhaopeng Tu, Haodong Duan. **Human Cognitive Benchmarks Reveal Foundational Visual Gaps in MLLMs**. arXiv:2502.16435. NeurIPS 2026. https://arxiv.org/abs/2502.16435

Keywords: speculative decoding • LLM inference acceleration • online distillation • chain-of-thought

Test-Time Speculation

Continuously Adapting the Draft Model During Generation Sustains Speculative Speedups Throughout Long Chain-of-Thought Outputs

By Avinash Kumar, Sujay Sanghavi, Poulami Das (UT Austin)

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Speculative decoding accelerates LLM inference by having a small draft model propose candidate tokens that a large target model verifies in parallel. In practice, however, the acceptance length of trained speculators—including state-of-the-art systems EAGLE-3, DFlash, and PARD—decays to near 1 after a few thousand output tokens, eliminating the speedup for long chain-of-thought generation. Test-Time Speculation (TTS) fixes this by adapting the draft model online, using the target-model evaluations that the verification step already performs, at zero added inference cost. On Qwen-3, Qwen-3.5, and Llama 3.1 model families, TTS improves acceptance length by up to 72% and 41% on average over the best static speculator, translating to 2.1–3.4× wall-clock speedup on 5K-token generation tasks.

AT A GLANCE

Problem: Trained speculators are distilled offline on short sequences; their acceptance length decays toward 1 on the long CoT outputs that matter most for reasoning tasks.

Why it matters: Speculative decoding is the leading inference acceleration technique, but its speedup evaporates precisely for the long-horizon tasks that consume the most compute.

Method: Online KL distillation of the draft model using target-model token probabilities already computed by the verification step, plus a proximity regularizer to prevent forgetting.

Result: Up to 72% improvement in acceptance length; 2.1–3.4× wall-clock speedup on 5K+ token outputs.

Evidence: MATH-500, HumanEval+, long-document summarization; Qwen-3/3.5 and Llama 3.1 families; ablations on regularizer and step count.

The Big Picture

Speculative decoding exploits the asymmetry between generation and verification: a small, fast draft model proposes k tokens, and the large target model verifies all k in one forward pass—accepting the longest prefix consistent with its distribution. When the draft model agrees with the target model, the effective throughput multiplies by k; when it does not, the cost is at most one wasted draft pass.

The technique's Achilles' heel is draft quality at long context lengths. All existing speculators are trained on datasets of typical generation lengths (hundreds to low thousands of tokens); at inference, reasoning-intensive tasks such as MATH, LiveCodeBench, and multi-step QA generate five to ten thousand tokens. The draft model's internal representations, tuned for short contexts, drift from the target's as the context grows, and acceptance length collapses.

Test-Time Speculation reframes speculative decoding as a continual learning problem: the draft model is a student that should track the target's distribution throughout generation, not just at the start.

Why Existing Approaches Fall Short

Static draft models (EAGLE-3, DFlash, PARD): All trained offline on sequences far shorter than inference-time CoT outputs. No mechanism exists to update the draft as the context lengthens.

Longer training sequences: Simply including longer sequences in offline training is expensive, requires representative CoT data that may not be available, and still does not adapt to the specific prompt or intermediate reasoning steps of a given inference call.

Online re-training approaches: Prior online adaptation methods for NLP require periodic re-training runs with additional target-model calls, breaking the generation pipeline and negating the speedup.

Core Mechanism

At each speculative decoding step, the target model already evaluates the k draft tokens and produces per-token probability distributions $\pi^*(\cdot \mid x_{<j})$ for $j = t, \ldots, t+k$. These distributions are the exact student-teacher signal needed to distill the draft model for the current context—and conventional speculative decoding discards them after acceptance/rejection decisions.

TTS re-uses these distributions to take one gradient step on the draft model after each verification pass. The gradient step minimizes a KL distillation loss plus a proximity regularizer that prevents the draft from forgetting its base distribution. Because the gradient is taken on a small draft model (1–3B parameters) and no additional target-model calls are needed, the per-step overhead is negligible.

Technical Formulation

Let π^* denote the target model and π_θ the draft model with parameters θ. At step t, the token-level KL distillation loss over the k verified positions is:

$$\mathcal{L}_t(\theta) = \frac{1}{k} \sum_{j=t}^{t+k} D_{\mathrm{KL}}(\pi^*(\cdot \mid x_{<j}) \parallel \pi_\theta(\cdot \mid x_{<j}))$$

The proximity regularizer prevents catastrophic forgetting:

$$\Omega(\theta, \theta_{t-1}) = \|\theta - \theta_{t-1}\|_2^2$$

The combined update is a single gradient step:

$$\theta_t = \theta_{t-1} - \eta \nabla_\theta [\mathcal{L}_t(\theta_{t-1}) + \beta \Omega(\theta_{t-1}, \theta_{t-2})]$$

with learning rate η following a cosine decay schedule over the generation horizon, and regularization weight β set so that total parameter shift remains within a small budget.

Learning or Inference Procedure

Standard speculative decoding (baseline): Frozen draft model; acceptance length decays with context length.

TTS (proposed):

1. Initialize π_θ from any pre-trained speculator.
2. At each verification step:
   (a) Accept/reject draft tokens as usual.
   (b) Compute $\mathcal{L}_t$ using the target distributions already obtained.
   (c) Take one gradient step to update θ.
3. Continue generation with the updated draft model.

No additional target-model calls. The gradient step on the small draft model adds $< 5\%$ latency overhead in practice.

What the Guarantee Says

The distillation loss minimized at each step is the expected log-probability ratio between the target and draft at the *current* context. Minimizing it pushes the draft toward the target's distribution at the positions just verified. The proximity regularizer bounds the per-step parameter shift, ensuring that adaptation at step t does not degrade draft quality at step $t-1$'s context distribution beyond a controlled amount. Together, these properties imply that the acceptance rate converges to a level bounded by the target's entropy at the current context—the theoretical maximum for any draft model—rather than decaying toward 1.

Experimental Findings

- **MATH-500 (long mathematical reasoning):** TTS achieves 3.4× wall-clock speedup over the target at 5K tokens; the best static speculator achieves 1.5× at the same length.
- **HumanEval+ (long code generation):** 2.7× vs. 1.8× for TTS and the best static speculator respectively.
- **Acceptance length trajectory:** Static speculators show a monotonic decay from ~ 4.5 tokens at 500 tokens to ~ 1.2 at 5K tokens; TTS maintains ~ 4.0 tokens throughout.
- **Model families:** Gains are consistent across Qwen-3-8B, Qwen-3.5-32B, and Llama-3.1-70B with their respective draft models.
- **Compatibility:** TTS improves acceptance length for all three base speculators (EAGLE-3, DFlash, PARD); EAGLE-3 + TTS achieves the best absolute acceptance length.

Ablations and Interpretation

No proximity regularizer: The draft overfits to the current context window and loses generalization to the next step; acceptance length drops by 18% compared to full TTS.

Multiple gradient steps per verification: Two steps per round adds compute without improving acceptance; one step is both necessary and sufficient.

Fixed vs. cosine learning rate: Cosine decay improves stability and final acceptance length by $\sim 5\%$; a fixed high rate causes oscillation in later generation steps.

Gains grow with output length: Improvement over the base speculator is near-zero at < 500 tokens (where the training distribution covers the context) and reaches its maximum at $> 5{,}000$ tokens, confirming that TTS specifically addresses the out-of-distribution long-context problem.

Reference

Avinash Kumar, Sujay Sanghavi, Poulami Das. **Test-Time Speculation**. arXiv:2605.09329. NeurIPS 2026. https://arxiv.org/abs/2605.09329

Keywords: safe reinforcement learning • constrained MDP • representation learning • low-rank MDP

Primal-Dual Representation Learning for Low-Rank Constrained MDPs

The First Sample-Efficient Algorithms with Theoretical Guarantees for Continuous-State Safe RL Under Both Soft and Hard Violation Constraints

By Chenhao Zhou, Chao Zhang, Wensong Bai, Hanbin Zhao, Hui Qian

NeurIPS 2026 (Poster #151600)

Safe reinforcement learning requires an agent to maximize return while keeping constraint violations within a budget—a challenge that, for continuous high-dimensional state spaces, has lacked end-to-end theoretical guarantees when the transition model must be learned from data. This paper introduces REP-PD-TEG and REP-PD-H-TEG, two primal-dual algorithms that jointly learn low-rank transition representations and satisfy safety constraints, achieving $\tilde{O}(\sqrt{K})$ return regret (soft violation) and certified per-episode constraint satisfaction (hard violation) respectively. The key insight is that representation learning and dual updates can be decoupled: the former proceeds by maximum-likelihood exploration, independent of the safety objective, while the latter handles constraints via an optimistic Lagrangian with a stabilized dual search.

AT A GLANCE

Problem: Safe RL in continuous state spaces requires learning transition representations and satisfying constraints simultaneously; no prior method provided end-to-end guarantees for both.

Why it matters: Real-world safe RL (robotics, autonomous driving) has continuous state spaces; tabular guarantees do not apply, and deep safe RL methods (CPO, PDO) lack theoretical backing.

Method: Decoupled representation learning via ML exploration, followed by optimistic Lagrangian planning (REP-PD-TEG for soft violation) or per-episode feasibility certification (REP-PD-H-TEG).

Result: $\tilde{O}(\sqrt{K})$ regret and $\tilde{O}(K^{3/4})$ total samples (soft); certified per-episode zero violation (hard).

Evidence: Continuous-state safety gridworld, constrained LQR; ablations on features, exploration rate, dual step size.

The Big Picture

Safe RL is the problem of training policies that maximize cumulative reward while respecting safety constraints expressed as budgets on cumulative or per-episode costs. It is central to deploying RL in physical systems—robots that should not fall, vehicles that should not collide, and medical dosing agents that should not exceed toxic thresholds.

The theoretical safe RL literature has largely studied tabular settings, where finite-sample guarantees can be derived from counting arguments over a discrete state space. For continuous state spaces, function approximation is necessary, and the low-rank MDP model—where the transition kernel factorizes as $P(s' \mid s, a) = \phi(s, a)^\top \mu(s')$ for low-dimensional features ϕ and μ—is the natural extension that retains mathematical tractability.

Until this paper, no work had provided end-to-end sample-efficiency guarantees for the combined problem of learning the representation ϕ and satisfying a safety constraint in a continuous-state low-rank CMDP.

Why Existing Approaches Fall Short

Tabular constrained RL (C-UCBVI, OptPess-LP): Provides finite-sample guarantees but requires discretizing the state space, which is computationally intractable in high dimensions and introduces approximation error proportional to the discretization resolution.

Linear function approximation for CMDPs: Assumes the true features ϕ are given; cannot handle the setting where they must be estimated from data, which introduces feature estimation error that can violate constraint guarantees.

Deep safe RL (CPO, PDO, RCPO): Achieves strong empirical performance but provides no finite-sample bounds on regret or constraint violation; convergence guarantees require idealized assumptions rarely met in practice.

Unconstrained low-rank MDP representation learning: Techniques such as FLAMBE and REPUCB learn ϕ by maximum likelihood and achieve provably near-optimal policies—but the dual variable of the safety constraint interacts non-trivially with the UCB bonus, breaking the direct extension.

Core Mechanism

The key architectural insight is that representation learning and safety constraint handling can be *decoupled*:

1. **Representation learning:** Run a decaying epsilon-greedy exploration policy to collect transition data; fit $\hat{\phi}$ by maximum-likelihood. This is independent of the safety objective.

2. **Primal-dual planning:** Use $\hat{\phi}$ to construct UCB-style optimistic estimates of both the return $\hat{V}_\pi^r$ and the constraint utility $\hat{V}_\pi^c$. Plan with the optimistic Lagrangian: $\max_\pi \min_\lambda \hat{V}_\pi^r - \lambda(\hat{V}_\pi^c - b)$. Update λ by dual gradient ascent.

REP-PD-TEG handles the soft (cumulative) violation case with standard UCB bonuses and gradient ascent on the dual.

REP-PD-H-TEG handles the hard (per-episode certified) violation case by adding a lower-confidence constraint certificate: before deploying any policy, the algorithm verifies that the pessimistic constraint estimate $\underline{V}_\pi^c$ is below the budget b, preventing episodes that violate the constraint even in the worst case.

Technical Formulation

The low-rank CMDP structure assumes:

$$P(s' \mid s, a) = \phi(s, a)^\top \mu(s'), \quad \phi : \mathcal{S} \times \mathcal{A} \to \mathbb{R}^d$$

The constrained optimization problem is:

$$\max_\pi V_\pi^r \quad \text{s.t.} \quad V_\pi^c \le b$$

The Lagrangian relaxation is:

$$\mathcal{L}(\pi, \lambda) = V_\pi^r - \lambda(V_\pi^c - b), \quad \lambda \ge 0$$

Maximum-likelihood feature learning from an epsilon-greedy exploration policy with budget T_0 achieves feature error $\|\hat{\phi} - \phi^*\|_\infty \le \epsilon_\phi$ with high probability for $T_0 = \tilde{O}(d^2/\epsilon_\phi^2)$.

With the learned features, the optimistic return and constraint estimates are:

$$\hat{V}_\pi^r = V_\pi^r(\hat{\phi}) + \underbrace{\beta_r \cdot b_\pi^\phi}_{\text{UCB bonus}}, \quad \hat{V}_\pi^c = V_\pi^c(\hat{\phi}) + \beta_c \cdot b_\pi^\phi$$

The main theoretical results for REP-PD-TEG are:

$$\text{Return regret} = \tilde{O}(\sqrt{K}), \quad \text{Total samples} = \tilde{O}(K^{3/4})$$

For REP-PD-H-TEG, the per-episode hard-violation guarantee holds with probability $1 - \delta$ for all K episodes simultaneously.

Learning or Inference Procedure

REP-PD-TEG:

1. Collect T_0 episodes of epsilon-greedy data; fit $\hat{\phi}$ by MLE.
2. For each episode k: compute optimistic $\hat{V}^r$, $\hat{V}^c$ using $\hat{\phi}$ and UCB bonuses; solve the optimistic Lagrangian for π_k; deploy π_k; update λ by gradient ascent: $\lambda \leftarrow \lambda + \eta_\lambda(V_{\pi_k}^c - b)$.

REP-PD-H-TEG:

1. Steps 1–2 as above, but before deploying π_k, compute the pessimistic lower bound $\underline{V}_{\pi_k}^c$ using a lower-confidence interval on $\hat{V}^c$.
2. If $\underline{V}_{\pi_k}^c \le b$: deploy π_k (certified safe).
3. Otherwise: invoke the stabilized dual search to find a feasible policy (shrink dual step size if the certificate is tight).

What the Guarantee Says

Soft violation (REP-PD-TEG): The cumulative return regret over K episodes is $\text{Reg}_K = \tilde{O}(\sqrt{K})$, and the total cumulative constraint violation is bounded by $\tilde{O}(\sqrt{K})$. Total episodes including exploration: $\tilde{O}(K^{3/4})$.

Hard violation (REP-PD-H-TEG): With probability $1 - \delta$, the deployed policy satisfies $V_{\pi_k}^c \le b$ for all $k = 1, \ldots, K$ simultaneously. Return regret is slightly worse than the soft case due to the conservative certification step but remains poly(K)-sublinear.

First result of its kind: The authors claim these are the first finite-sample guarantees for representation learning in low-rank CMDPs with continuous state spaces under both soft and certified hard violation settings.

Experimental Findings

- **Continuous safety gridworld:** REP-PD-TEG keeps cumulative constraint violations below budget; REP-PD-H-TEG achieves zero per-episode violations after an initial feature-learning phase.
- **Constrained LQR:** Both algorithms outperform CPO and PDO (which use fixed features) when the provided features are incorrect; performance matches CPO with correct features.
- **Benefit of learned features:** Methods with fixed (incorrect) features consistently violate the constraint budget; the adaptive feature learning of REP-PD-TEG removes this failure mode.

Ablations and Interpretation

Fixed vs. learned features: Ground-truth features in the primal-dual step achieve near-optimal performance; incorrect fixed features fail to satisfy constraints and achieve poor return, confirming the necessity of end-to-end representation learning.

Epsilon-greedy schedule: A decaying schedule is critical; a constant rate leads to over-exploration and slower convergence; an overly fast decay leads to poor feature quality and constraint violations.

Dual step size stability (REP-PD-H-TEG): The stabilized dual search (reduced step size near the con-

straint boundary) is essential for hard-violation safety; naive gradient ascent on the dual oscillates and produces intermittent per-episode violations.

Reference

Chenhao Zhou, Chao Zhang, Wensong Bai, Hanbin Zhao, Hui Qian. **Primal-Dual Representation Learning for Low-Rank Constrained MDPs**. NeurIPS 2026 (Poster #151600). https://neurips.cc/virtual/2026/poster/151600